MODULE 1

MATHEMATICAL LOGIC

Solved Previous-Year Questions • Model Questions • Short Explanations

Course: 3341 Discrete Mathematics

For: Diploma students in Artificial Intelligence and Machine Learning

Topics: Propositional Logic, Connectives and Truth Tables, Implication, Tautology, Contradiction, Contingency, Converse/Inverse/Contrapositive, Duality, Predicate Logic, WFF, Quantifiers, Natural Deduction, Rules of Inference, Methods of Proof, and Resolution Principle.

How to use this booklet

The first section solves the readable logic questions in the shared handwritten Module 1 pages. The second section provides new model questions in similar styles. Each solution includes a short explanation.

Scope note: One visible question asks for the number of subsets of a set, and an earlier image asks about the Binomial Theorem. Those topics are outside the Mathematical Logic syllabus listed in the request, so they are not included in the main solved set.

Quick concept recap

• Proposition: a declarative statement that is either true or false.

• ¬p means NOT p; $p \wedge q$ means p AND q; $p \vee q$ means p OR q.

• $p \rightarrow q$ is false only when p is true and q is false.

• Tautology: always true. Contradiction: always false. Contingency: true for some assignments and false for others.

• Converse of $p \rightarrow q$: $q \rightarrow p$. Inverse: $\neg p \rightarrow \neg q$. Contrapositive: $\neg q \rightarrow \neg p$.

• Original implication and contrapositive are logically equivalent. Converse and inverse are equivalent to each other.

• WFF means the expression follows the syntax rules of logic.

SECTION A — SOLVED PREVIOUS-YEAR QUESTIONS

The answers below are based on the supplied question images. Where a handwritten step is compressed, the reasoning is expanded for clarity.

PYQ 1. Double negation [1 mark]

Question: If p is a proposition, simplify ¬(¬p).

Solution:

```
¬(¬p) ≡ p
```

Answer: **p**.

Explanation: Negating twice restores the original truth value.

PYQ 2. “I pass only if you pass” [1 mark]

Question: Let p = “I pass” and q = “you pass”. Express the statement logically.

Solution:

```
p → q
```

Equivalent form: **¬p ∨ q**.

Contrapositive: ¬q → ¬p — “If you fail, then I fail.”

Explanation: “p only if q” means q is necessary for p, so p → q.

PYQ 3. Dual of p ↔ q [1 mark]

Question: Write the dual of p ↔ q using the expansion in the handwritten solution.

Solution:

```
p ↔ q ≡ (p → q) ∧ (q → p)
≡ (¬p ∨ q) ∧ (¬q ∨ p)
Dual: (¬p ∧ q) ∨ (¬q ∧ p)
```

Answer: **(¬p ∧ q) ∨ (¬q ∧ p)**.

Explanation: Interchange ∧ and ∨; keep variables and negations unchanged.

PYQ 4. Truth table for conjunction [3 marks]

Question: Determine the truth table of p ∧ q.

Solution:

p ∧ q is true only when both p and q are true.

p	q	p ∧ q
T	T	T
T	F	F
F	T	F
F	F	F

Explanation: The word AND requires both statements to be true.

PYQ 5. Show that ¬(p → q) → ¬q is a tautology [3 marks]

Question: Use a truth table.

Solution:

Calculate p → q first, then its negation, then ¬q, then the final implication.

p	q	$p \to q$	$\neg(p \to q)$	$\neg q$	Final
T	T	T	F	F	T
T	F	F	T	T	T
F	T	T	F	F	T
F	F	T	F	T	T

Every value in the final column is T. Therefore, the formula is a tautology.

PYQ 6. Prove without a truth table [3 marks]

Question: Prove $(p \wedge q) \to (p \to q)$.

Solution:

1. $p \wedge q$ Premise
2. q From 1, Simplification
3. $\neg p \vee q$ From 2, Addition
4. $p \to q$ Since $p \to q \equiv \neg p \vee q$

Therefore $(p \wedge q) \to (p \to q)$.

Explanation: The premise gives q. Hence $\neg p \vee q$ is true, which is equivalent to $p \to q$.

PYQ 7. Truth table for $\neg(p \vee q) \to \neg p$ [7 marks]

Question: Construct the truth table.

Solution:

Calculate $p \vee q$, then $\neg(p \vee q)$, then $\neg p$.

p	q	$p \vee q$	$\neg(p \vee q)$	$\neg p$	Final
T	T	T	F	F	T
T	F	T	F	F	T
F	T	T	F	T	T
F	F	F	T	T	T

The final column is all T, so this formula is a tautology.

PYQ 8. Validity by method of proof [7 marks]

Question: Premises: $(\neg p \vee q) \to r$; $r \to (s \vee t)$; $\neg s \wedge \neg u$; $\neg u \to \neg t$. Conclusion: $\therefore p$.

Solution:

1. $\neg s \wedge \neg u$ Premise
2. $\neg u$ Simplification from 1
3. $\neg u \to \neg t$ Premise
4. $\neg t$ Modus Ponens from 2, 3
5. $\neg s$ Simplification from 1
6. $\neg s \wedge \neg t$ Conjunction from 4, 5
7. $r \to (s \vee t)$ Premise
8. $\neg(s \vee t) \to \neg r$ Contrapositive of 7
9. $\neg(s \vee t)$ De Morgan’s law from 6
10. $\neg r$ Modus Ponens from 8, 9
11. $(\neg p \vee q) \to r$ Premise

12. ¬(¬p ∨ q) Modus Tollens from 11, 10

13. p ∧ ¬q De Morgan's law and double negation

14. p Simplification from 13

Therefore, the argument is valid.

Explanation: ¬s and ¬t make s ∨ t false, forcing r to be false. The first implication then forces ¬p ∨ q to be false, so p is true.

PYQ 9. Any seven rules of inference [7 marks]

Question: List seven rules with their symbolic forms.

Solution:

1. Modus Ponens: p, p → q ■ q.

2. Modus Tollens: p → q, ¬q ■ ¬p.

3. Hypothetical Syllogism: p → q, q → r ■ p → r.

4. Disjunctive Syllogism: p ∨ q, ¬p ■ q.

5. Conjunction: p, q ■ p ∧ q.

6. Simplification: p ∧ q ■ p (or q).

7. Addition: p ■ p ∨ q.

Explanation: A rule of inference is a valid pattern for deriving a conclusion from premises.

PYQ 10. Converse, inverse and contrapositive [7 marks]

Question: For "If –1 < 3 and 3 + 7 = 10, then sin(3π/2) = –1", write the converse, inverse, contrapositive and truth values.

Solution:

Let p = [–1 < 3 and 3 + 7 = 10], q = [sin(3π/2) = –1]. Both p and q are true.

Original: p → q = T → T = True.

Converse: q → p = T → T = **True**.

Inverse: ¬p → ¬q = F → F = **True**.

Contrapositive: ¬q → ¬p = F → F = **True**.

Explanation: An implication with a false antecedent is true. Here p and q are both true.

SECTION B — MODEL QUESTIONS WITH ANSWERS

These are new practice questions based on the style and marks visible in the shared pages. They are model questions, not actual past-paper questions.

PART A — 1 mark each

Model A1. Proposition [1 mark]

Question: Define a proposition.

Solution:

A proposition is a declarative statement that is either true or false, but not both.

Model A2. Conjunction [1 mark]

Question: When is $p \wedge q$ true?

Solution:

Only when both p and q are true.

Model A3. Implication [1 mark]

Question: Write $p \rightarrow q$ using $\neg$ and $\vee$.

Solution:

$$p \rightarrow q \equiv \neg p \vee q$$

Explanation: Both expressions have the same truth values.

Model A4. Classification [1 mark]

Question: Classify $p \vee \neg p$.

Solution:

It is a **tautology**.

Explanation: Either p is true or its negation is true.

Model A5. Quantifiers [1 mark]

Question: What do $\forall$ and $\exists$ mean?

Solution:

$\forall$ means “for all”; $\exists$ means “there exists”.

PART B — 3 marks each

Model B1. Truth table for implication [3 marks]

Question: Construct the truth table for $p \rightarrow q$.

Solution:

The implication is false only when p = T and q = F.

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

When p is false, the implication is true regardless of q.

Model B2. Converse, inverse and contrapositive [3 marks]

Question: For “If a number is divisible by 4, then it is even”, write the three forms.

Solution:

Let p = divisible by 4, q = even.

Converse: $q \to p$ — If a number is even, then it is divisible by 4.

Inverse: $\neg p \to \neg q$ — If it is not divisible by 4, then it is not even.

Contrapositive: $\neg q \to \neg p$ — If it is not even, then it is not divisible by 4.

Explanation: The contrapositive is equivalent to the original. The converse is not generally equivalent.

Model B3. Well-formed formula [3 marks]

Question: Say whether these are WFFs: (i) $(p \wedge q) \to r$, (ii) $p \wedge \to q$, (iii) $\neg(p \vee q)$.

Solution:

(i) **WFF** — correctly formed.

(ii) **Not a WFF** — the connectives are incorrectly arranged.

(iii) **WFF** — a valid disjunction is negated.

Explanation: A WFF follows the syntax rules of logic.

Model B4. Classify formulas [3 marks]

Question: Classify (i) $p \wedge \neg p$, (ii) $p \vee \neg p$, (iii) $p \to q$.

Solution:

(i) **Contradiction**.

(ii) **Tautology**.

(iii) **Contingency**.

Explanation: Use the final truth-table column: all F, all T, or a mixture.

Model B5. Natural deduction [3 marks]

Question: From $p \wedge q$, derive q and name the rule.

Solution:

```
1. p ∧ q Premise
2. q Simplification (∧-elimination)
```

Explanation: A conjunction lets us take either component separately.

PART C — 7 marks each

Model C1. Compound truth table [7 marks]

Question: Construct a truth table for ¬(p ∨ q) → ¬p.

Solution:

Find p ∨ q, then ¬(p ∨ q), then ¬p, then the implication.

p	q	p ∨ q	¬(p ∨ q)	¬p	Final
T	T	T	F	F	T
T	F	T	F	F	T
F	T	T	F	T	T
F	F	F	T	T	T

The final column is all T; therefore the formula is a tautology.

Model C2. Rules of inference [7 marks]

Question: Using inference rules, prove q from p → q and p.

Solution:

```
1. p → q Premise
2. p Premise
3. q Modus Ponens from 1 and 2
```

Explanation: Modus Ponens has the form p → q, p ■ q.

Model C3. Proof by contrapositive [7 marks]

Question: Prove: If n^2 is even, then n is even.

Solution:

Contrapositive: If n is odd, then n^2 is odd.

Let n = 2k + 1 for an integer k.

```
n² = (2k + 1)² = 4k² + 4k + 1
n² = 2(2k² + 2k) + 1
```

Thus n^2 is of the form 2m + 1, so it is odd. The contrapositive is true; therefore the original statement is true.

Explanation: A statement and its contrapositive are logically equivalent.

Model C4. Resolution principle [7 marks]

Question: Use resolution to show p follows from p ∨ q and ¬q.

Solution:

Clauses: (1) p ∨ q; (2) ¬q.

```
(p ∨ q), ¬q ■ p
```

The complementary literals q and ¬q cancel, leaving p.

Explanation: Resolution combines clauses containing complementary literals.

Model C5. Predicate logic and quantifiers [7 marks]

Question: Let the domain be all integers. Express “Every integer has a successor.”

Solution:

```
∀x ∈ ■, ∃y ∈ ■ such that y = x + 1.
```

Equivalently: ∀x ∃y (y = x + 1), with the domain specified as integers.

Explanation: ∀ applies to every integer x; ∃ is satisfied by choosing y = x + 1.

SECTION C — EXTRA PRACTICE FOR FULL SYLLABUS COVERAGE

The visible PYQ pages focus mainly on propositional logic, truth tables, implication transformations and inference. These short questions cover additional topics listed in the syllabus.

Predicate logic

Question: What is a predicate?

Answer: A statement with one or more variables that becomes a proposition when values are assigned or variables are quantified.

Universal quantifier

Question: Translate “Every student studies logic.”

Answer: $\forall x$ (Student(x) $\rightarrow$ StudiesLogic(x)).

Existential quantifier

Question: Translate “Some student passed.”

Answer: $\exists x$ (Student(x) $\wedge$ Passed(x)).

Duality principle

Question: How is the dual formed?

Answer: Interchange $\wedge$ and $\vee$; if T and F occur, interchange them too. Keep variables and negation signs unchanged.

Natural deduction

Question: What is natural deduction?

Answer: A proof method that derives a conclusion from premises by applying valid inference rules step by step.

Methods of proof

Question: Name four methods of proof.

Answer: Direct proof, contrapositive, contradiction, and proof by cases.

Resolution

Question: State the resolution rule.

Answer: (A $\vee$ B) and (¬B $\vee$ C) imply (A $\vee$ C).

Logical equivalence

Question: What does p $\equiv$ q mean?

Answer: p and q have the same truth value under every truth assignment.

Biconditional

Question: When is p $\leftrightarrow$ q true?

Answer: When p and q have the same truth value: both true or both false.

Exam tips and common mistakes

• For n distinct propositions, a truth table has 2■ rows. Two propositions need 4 rows; three propositions need 8.

• Work out intermediate columns in bracket order and finish with the main connective.

• For p $\rightarrow$ q, only T $\rightarrow$ F gives F.

• Original: p $\rightarrow$ q; converse: q $\rightarrow$ p; inverse: ¬p $\rightarrow$ ¬q; contrapositive: ¬q $\rightarrow$ ¬p.

• In proof questions, write each statement with its reason. Marks are often awarded for the steps and rule names.

• All T in the final column = tautology; all F = contradiction; a mixture = contingency.

Mark pattern

The handwritten pages show 1-mark short questions, 3-mark truth-table/short-proof questions, and 7-mark extended truth-table/proof questions. The model section follows that visible pattern. Check the official current exam scheme if the mark allocation has changed.

End of Module 1 — Mathematical Logic